

Greedy Algorithm

① Optimal solⁿ (max, min)
↳ Sorting ✓

② No fixed rule

③ lots of time global optimum solⁿ is not achieved.

Greedy Algo is the problem solving technique where we make the locally optimal choices ^(best) at each stage and try to achieve global optimum solⁿ.

Question:

Min Absolute difference pair:

Given 2 array A and B of same size. pair $(i, j) \mid i \in A, j \in A$
Sum = $\sum |i - j|$, find min. sum.

any combination of element of A, B
↓

$$A = [1, 2, 3]$$

$$B = [2, 1, 3]$$

Can 1
 $(1, 2) \quad (2, 1) \quad (3, 3) \rightarrow \text{sum} = 1 + 1 + 0 = 2$

Can 2
 $(1, 1), (2, 2), (3, 3) \rightarrow \text{sum} = 0 + 0 + 0 = 0 \text{ (min)}$

↳ combinations

Can 3
 $(1, 3) \quad (2, 1) \quad (3, 1) \rightarrow \text{sum} = 2 + 1 + 1 = 4$

⋮

$$\text{sum} = 0$$

for $(i = 0 \text{ to } n) \{$

$$\text{sum} += \text{Math.abs}(A[i] - B[i]);$$

$\}$

ans sum

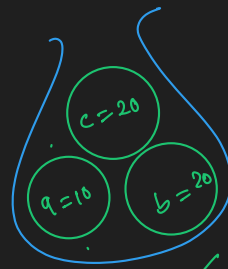
$$A \rightarrow \text{Sort} \quad [1, 2, 3]$$

$$B \rightarrow \text{Sort} \quad [1, 2, 3]$$

$$P/w = \begin{bmatrix} 6 & 5 & 4 \\ a & b & c \end{bmatrix}$$

↓
desc order

Product	a	b	c
P/w	6	5	4



$$W = 50 \text{ } \cancel{10} \text{ } \cancel{20}$$

$$\text{Profit} = 6 \times 10 = 60 \\ + 5 \times 20 = 100 \\ + 4 \times 20 = 80$$

$$Wt[b] \leq W?$$

if true then take complete wt

$$Wt[c] \leq W?$$

no, then take the remain W of knapsack for the item 'c'

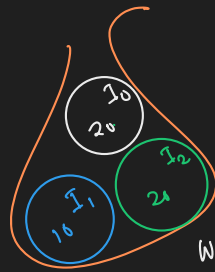
$$Wt = [30 \ 10 \ 20] \\ \text{Price} = [120 \ 60 \ 100]$$

map

item	wt	P/w
0	30	4
1	10	6
2	20	5

sort based on P/w in desc

item	wt	P/w
1	10	6
2	20	5
0	30	4



$$W = 50 \text{ } \cancel{10} \text{ } \cancel{20}$$

$$\text{Profit} = 6 \times 10 + 5 \times 20 + 4 \times 20 \\ = 60 + 100 + 80 \\ = 240$$

Question 4

Activity Selection

$A_0 \ A_1 \ A_2 \ A_3 \ A_4 \ A_5$

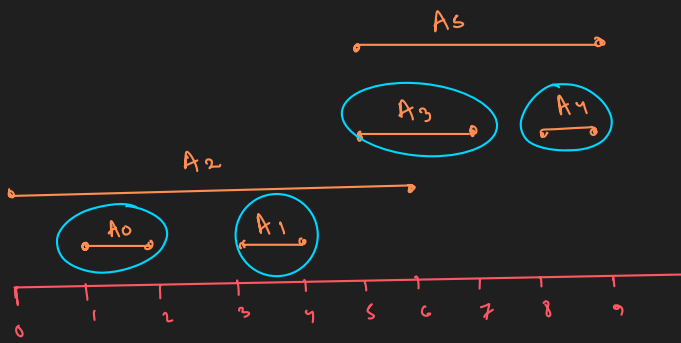
$$\text{Start time} = [1, 3, 0, 5, 8, 5]$$

$$W_{(\text{sort})} \text{ end time} = [2, 4, 6, 7, 9, 9]$$

$$A_{\text{ans}} = 4$$

Given all activities are sorted according to the end time

no. of max active perform w/o overlapping and print the activity.



non-overlapping activity

start time γ , end time of previous activity

input above sorting is not there based on end time.

$$s.t = [0, 1, 3, 5, 5, 8]$$

$$e.t = [6, 2, 4, 7, 9, 9]$$

$$Ans = A_1, A_2, A_3, A_5$$

Question 5

for each pair, (a, b) $a < b$ and you have to form a chain from those given pair st. (a, b) (c, d) iff $b < c$ ^{largest}

$$pairs = \checkmark (5, 24)$$

$$(30, 60)$$

$$(5, 28)$$

$$\checkmark (27, 40)$$

$$\checkmark (50, 90)$$

pair 2 included

$$curr\ st > prev\ end\ value$$

Job Sequencing Problem

job $\rightarrow$ (deadline, profit)

$$J_0 \quad 2, 80 \checkmark$$

$$J_1 \quad 6, 65 \checkmark$$

$$J_2 \quad 6, 70 \checkmark$$

$$J_3 \quad 5, 60 \checkmark$$

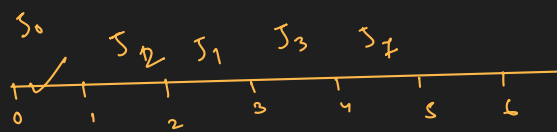
$$J_4 \quad 2, 22 \times$$

$$J_5 \quad 9, 25 \times$$

$$J_6 \quad 2, 10 \times$$

$$J_7 \quad 9, 20 \checkmark$$

max. profit $\uparrow$, at a time one job.
 $\hookrightarrow$ each job takes 1 unit of time



$$80 + 70 + 65 + 60 + 20$$

$$= 295$$

Ans 2
322

Out of Scope

Order

job 1 (profit) - job 2

job 1 profit - job 2 profit

job 1 profit - job 2 profit

job 2 (2, 80) | J1 (6, 70)

job 1 (6, 70) | J2 (2, 80)

2 < 3 < 4

4 > 3 > 2

Comparison → result

+ve → order maintain

-ve → order doesn't maintain

	deadline	profit
J ₀	2	80
J ₁	6	65
J ₂	6	70
J ₃	5	60
J ₄	2	22
J ₅	4	25
J ₆	2	10
J ₇	4	20



Sort (profit)

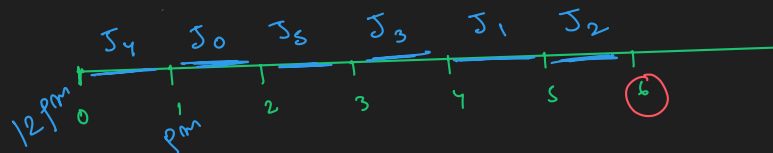
H.W
f.c analysis

After Sorting

J ₀	2	80
J ₂	6	70
J ₁	6	65
J ₃	5	60
J ₅	4	25
J ₄	2	22
J ₇	4	20
J ₆	2	10

Profit = 322

max deadline = 6



Array [max deadline + 1] =

-1	4	0	5	3	1	2
0	1	2	3	4	5	6